

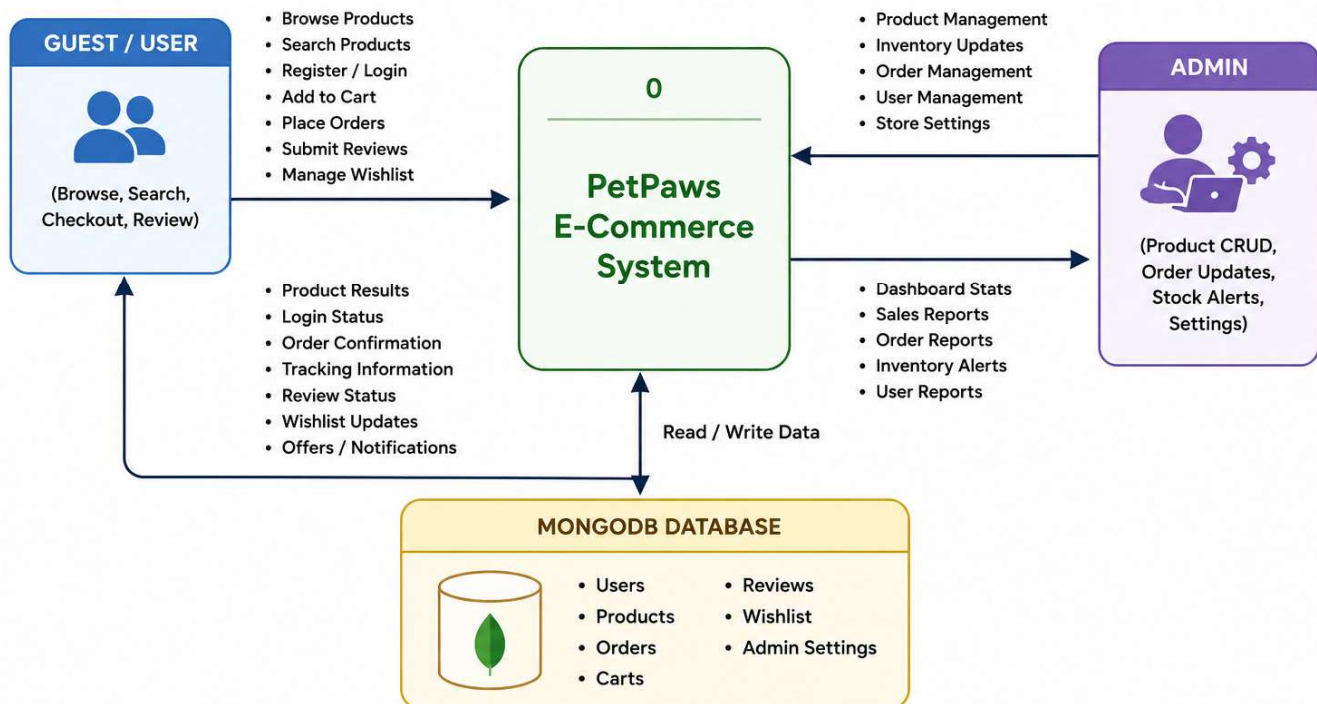
DOCUMENT: DATA FLOW DIAGRAMS & USER STORIES

Phase	Project Design Phase II
Document	Data Flow Diagram & User Stories
Project Name	PetPaws – Pet Foods & Accessories Store
Team ID	[TEAM-ID]
Date	14 June 2026
Maximum Marks	4 Marks

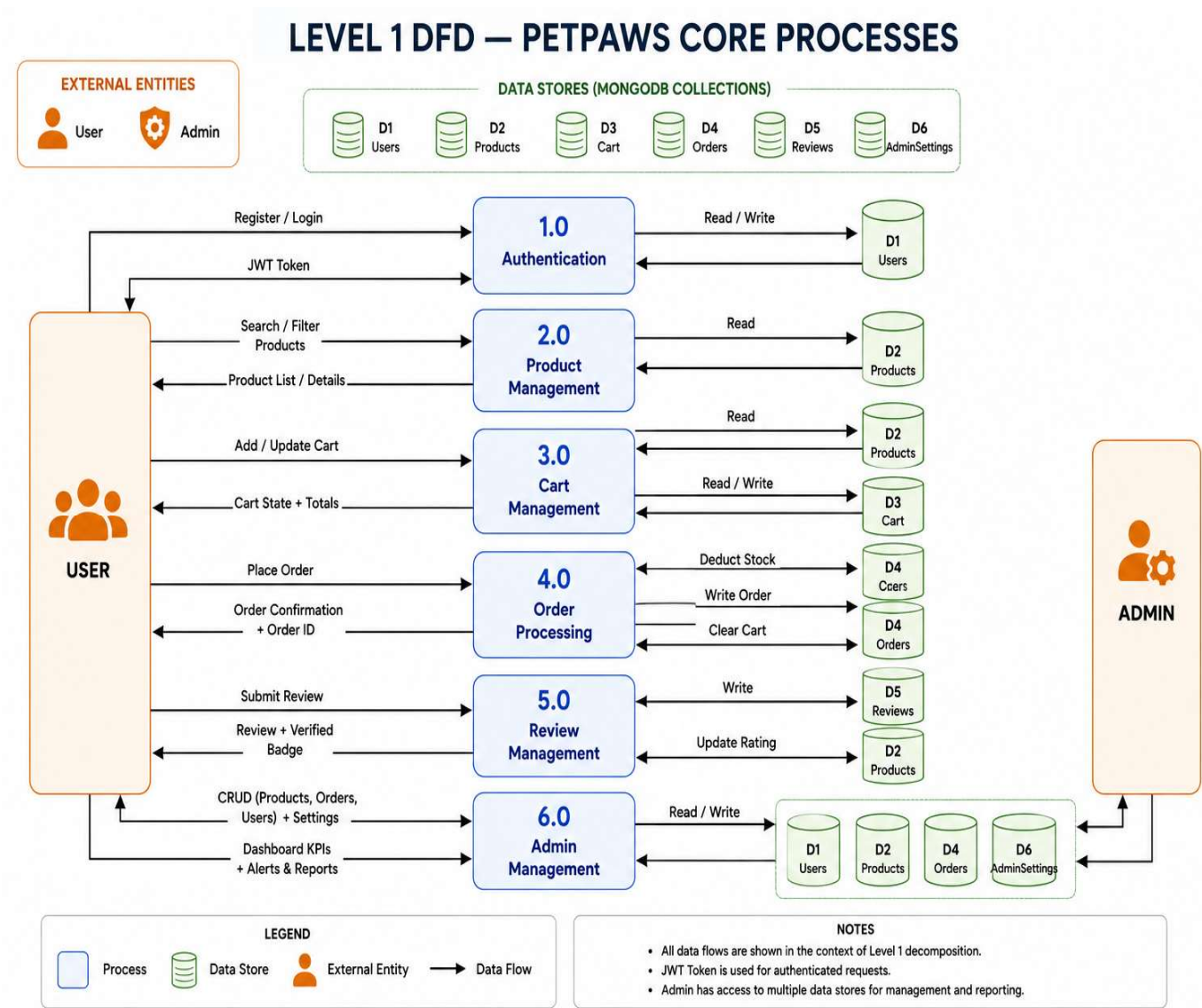
Data Flow Diagrams (DFD)

Level 0 DFD — Context Diagram

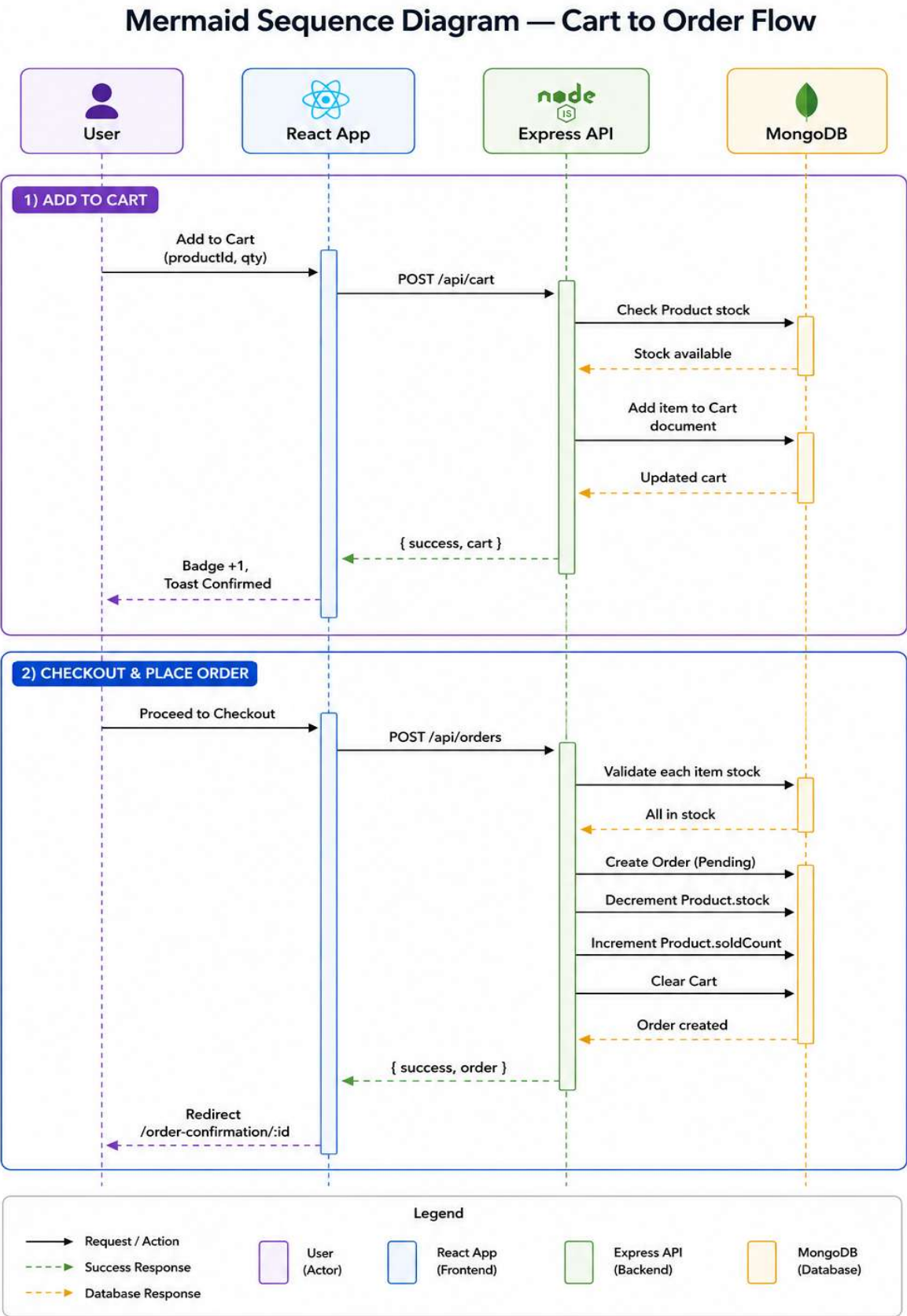
LEVEL 0 DFD — CONTEXT DIAGRAM PetPaws E-Commerce System



Level 1 DFD — Decomposed Processes



Mermaid Sequence Diagram — Cart to Order Flow



User Stories

User Type	Epic	USN	User Story	Acceptance Criteria	Priority	Release
Customer	Registration & Login	USN-1	As a user, I can register by entering my username, email, and password	Account is created; JWT returned; user redirected to homepage	High	Sprint-1
		USN-2	As a user, I can log in with email and password and receive a JWT token	JWT stored in localStorage; AuthContext updated; redirect to previous page	High	Sprint-1
	Product Browsing	USN-3	As a user, I can browse all pet products with category and pet-type filters	Product grid renders with applied filters; total count displayed; pagination works	High	Sprint-1
		USN-4	As a user, I can search products by keyword and sort by price/rating/newest	Results update on search/sort; empty state shown when no results	High	Sprint-1
		USN-5	As a user, I can view a product detail page with images, variants, and reviews	Image gallery renders; variant buttons appear; reviews tab shows ratings	High	Sprint-1
	Cart & Wishlist	USN-6	As a user, I can add products to my cart and see a real-time cart badge	Cart badge increments; item persisted to MongoDB; toast confirmation shown	High	Sprint-1
		USN-7	As a user, I can add products to a wishlist and move them to the cart	Wishlist page shows saved products; move-to-cart clears item from wishlist	Medium	Sprint-2
	Checkout	USN-8	As a user, I can complete a 3-step checkout: address, payment, review	All 3 steps navigable; order placed on Step 3 confirm; redirected to confirmation page	High	Sprint-2
		USN-9	As a user, I can select saved	Saved addresses shown as quick-	High	Sprint-2

			addresses or enter a new address at checkout	select buttons; new address form validates all required fields		
	Order Management	USN-10	As a user, I can view my order history and filter by status	Order list renders with status badges; filter tabs change displayed orders	High	Sprint-2
		USN-11	As a user, I can track my order status via a visual timeline and cancel eligible orders	5-stage timeline shown; Cancel button visible only for Pending/Confirmed; stock restored on cancel	High	Sprint-2
	Reviews & Profile	USN-12	As a user, I can submit a star rating and review for a product I purchased	Review appears on product page; Verified Purchase badge shown for delivered orders; rating average updates	Medium	Sprint-3
		USN-13	As a user, I can update my profile, change password, and manage addresses	Profile tab saves name/phone; password tab validates current password; address CRUD works correctly	Medium	Sprint-3
Administrator	Dashboard	USN-14	As an admin, I can view KPI cards, revenue chart, low-stock alerts, and top products	Dashboard renders 4 KPI cards; bar chart shows monthly revenue; low-stock table shows items < 10 units	High	Sprint-3
	Product CRUD	USN-15	As an admin, I can create, edit, and soft-delete pet products	Product appears in listing after creation; edit form pre-populates; soft-delete hides product from customers	High	Sprint-1
	Order Management	USN-16	As an admin, I can update order status and add tracking numbers	Status dropdown updates order; tracking number visible to customer;	High	Sprint-3

				statusHistory entry created		
	Customer Management	USN-17	As an admin, I can search customers and activate/deactivate accounts	Search returns filtered results; toggle changes isActive; deactivated user cannot log in	Medium	Sprint-4
	Store Settings	USN-18	As an admin, I can configure shipping fee, tax rate, and homepage banners	Settings saved and reflected immediately in checkout totals and homepage carousel	Medium	Sprint-4